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VANDERBILT UNIVERSITY

ME 171 INSTRUMENTATION LABORATORY

LAB 8: MEASUREMENT UNCERTAINTY

The results of this experiment will not be written in the form of a lab report. Instead, you will complete this lab and hand in all material to be graded at the end of the lab section. Therefore, write neatly on the pages attached – the TA grading your lab should be able to follow what you are doing. Remember to include units.

The objective of this experiment is to determine the fastest, most cost effective method of measuring the volume of a metal object while at the same time keeping the uncertainty of the volume measurement within a desired range.

Imagine this scenario: You are employed by a manufacturer who makes a product that is a boxlike piece of metal. For some unknown but important reason the volume of each piece being produced must be measured before being shipped out. You are determining the measurement procedures that others will use when they measure the pieces. The specifications that your supervisor has relayed to you are these:

1. The volume of each piece is determined by three length measurements.
2. Each of these three measurements can be made with either dial calipers or a micrometer.
3. It takes a longer time for a worker to measure a length with a micrometer than with dial calipers, therefore it is most desirable for the three measurements to be made with the dial calipers. However:
4. The uncertainty in the measured volume must be no greater than 5% (90% confidence) of the measured volume. This requirement must be met.

Keeping this scenario in mind, perform the following tasks in the lab:

1) Examine the dial calipers and micrometer. Determine what the input range and the least count are for each. Record these numbers here:

	Dial Calipers	Micrometer
Input Range,	0 to 6 inches	0.0001in to 1 in

Least Count, inches	.001 in	.0001in
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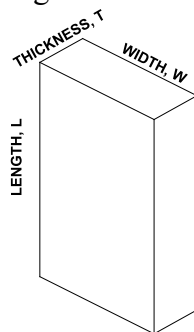
The readability for each instrument is somewhat subjective, but should be equal to or less than the least count for each instrument. To make it easy on the T.A. who will grade your work, let's assume that the readability for the dial calipers is one half of its least count (meaning that you can tell if it is in between two lines), while the readability for the micrometer is its least count. Write these numbers below, and calculate the dynamic range for each instrument (the **dynamic range** is the input range divided by the readability, and represents the number of discernable divisions of the input range the measurement device gives).

	Dial Calipers	Micrometer
Readability, inches	0.005	.0001
Dynamic Range, dimensionless	____1,200____	10,000

Let's assume that the uncertainty (with 90% confidence) in a measurement from either instrument is +/- the readability of the instrument. Write down the uncertainty for each instrument below:

Dial Calipers	Micrometer
Uncertainty, inches +/- 0.005 (90% confidence)	+/- 0.001 (90% confidence)

2) The part whose volume is to be measured is the 0.050 inch thick gage block. Take the 0.050 inch gage block and measure its three sides, using the dial calipers. These three sides are the length (L) which is the longest side, the width (W) which is the second longest side, and the thickness (T) which is the smallest side. The figure below shows this:



Measure the length, width and thickness of the block using the dial calipers (make sure that when the calipers are closed, the dial reads zero – achieve this by adjusting the faceplate):

L	1.8	inches +/- .005 inches (90% confidence)
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W .35 inches +/- .005 inches (90% confidence)

T .05 inches +/- .005 inches (90% confidence)

Now measure the width and thickness using the micrometer. Notice that the length is outside of the input range of the micrometer. Record the data below:

Micrometer Measurements:

W 0.3494 inches +/- .0001 inches (90% confidence)

T 0.0499 inches +/- .0001 inches (90% confidence)

You will also notice that on many of the micrometers zero width does not read 0.0000 inches. Record the zero offset here and make sure to account for it to get the real width (W) and length (L). **Also make sure to propagate the uncertainty in this measurement!**

Zero offset 0 inches +/- .0001 inches (90% confidence)

Note: uncertainty in the zero position exists for all three measurements for both the micrometer and the calipers. We will name these zero offsets and their uncertainties: L_0 , W_0 , T_0 and σ_{L_0} ,

σ_{W_0} , σ_{T_0} respectively.

3) Now, calculate the volume, V, of the block by using the three dial caliper measurements. Also calculate the uncertainty in the volume measurement, σ_V . Recall from lecture that if a derived (or calculated) measurement R comes from n different primary measurands, x_i :

Volume from calipers: $.0315 \text{ in}^3$

$$R = R(x_1, x_2, \dots, x_n) \quad (1)$$

then, the uncertainty in R , σ_R , can be calculated as:

$$\sigma_R = \left[\left(\frac{\partial R}{\partial x_1} \right)^2 \sigma_1^2 + \left(\frac{\partial R}{\partial x_2} \right)^2 \sigma_2^2 + \dots + \left(\frac{\partial R}{\partial x_n} \right)^2 \sigma_n^2 \right]^{\frac{1}{2}} \quad (2)$$

where $(\partial R / \partial x_i)$ is the partial derivative of R with respect to x_i , and σ_i is the uncertainty in x_i .

This is called the Kline and McClintock method, named after the two who developed this procedure back in 1953.

Specifically for this problem you need to calculate $(\partial V/\partial L)$, $(\partial V/\partial L_0)$, $(\partial V/\partial W)$, $(\partial V/\partial W_0)$, $(\partial V/\partial T)$ and $(\partial V/\partial T_0)$. First, determine the theoretical (or analytical) form of each partial derivative:

$$\text{For } V = (L-L_0)(W-W_0)(T-T_0)$$

$$\partial V/\partial L = (W-W_0)(T-T_0)$$

$$\partial V/\partial L_0 = -(W-W_0)(T-T_0)$$

$$\partial V/\partial W = (L-L_0)(T-T_0)$$

$$\partial V/\partial W_0 = -(L-L_0)(T-T_0)$$

$$\partial V/\partial T = (L-L_0)(W-W_0)$$

$$\partial V/\partial T_0 = -(L-L_0)(W-W_0)$$

Now using the measurements from the **dial calipers**, determine the numerical values for these partial derivatives:

$$\underline{\partial V/\partial L = .35(.005) = .00175 \text{ in}^2}$$

$$\underline{\partial V/\partial L_0 = -(.35)(.005) = -.00175 \text{ in}^2}$$

$$\underline{\partial V/\partial W = 1.8(.005) = .009 \text{ in}^2}$$

$$\underline{\partial V / \partial W = -1.8(.05) = -.009 \text{ in}^2}$$

$$\underline{\partial V / \partial T = 1.8(.35) = .63 \text{ in}^2}$$

$$\underline{\partial V / \partial T_0 = -1.8(.35) = -.63 \text{ in}^2}$$

(L, T, W: Caliper) Now in the space below determine σ_V using the numerical values for the partial derivatives and the uncertainties using the **dial calipers** for all primary measurements. Make sure to **clearly show** the formula and then its numerical evaluation.

$$\sigma_v = \sqrt{2\left(\frac{\partial V}{\partial L}\right)^2 \sigma^2 + 2\left(\frac{\partial V}{\partial T}\right)^2 \sigma^2 + 2\left(\frac{\partial V}{\partial W}\right)^2 \sigma^2}$$

$$\sigma_v = \sqrt{2(3.494 \cdot 10^{-5} \text{ in})^2 (.0001 \text{ in})^2 + 2(0.63 \text{ in})^2 (.005 \text{ in.})^2 + 2(0.00175 \text{ in})^2 (.005 \text{ in})^2}$$

$$= .00454 \text{ in}^3$$

Does this satisfy the criterion that the derived measurement uncertainty be 5% or less of the derived measurement?

$$\frac{.00454}{.0315} = .14142(100\%) = 14.142\%$$

Ans: no

(L:caliper), (W,T: micrometer) Repeat the above calculation for σ_V , now using the uncertainties for the micrometer when it is used to measure both W and T (and where L is still measured using the calipers). You will need to recalculate the numerical values for the partial derivatives – show these. Then make sure to **clearly show** the formula for σ_V and then its numerical evaluation.

$$\sigma_v = \sqrt{2\left(\frac{\partial V}{\partial L}\right)^2 \sigma^2 + 2\left(\frac{\partial V}{\partial T}\right)^2 \sigma^2 + 2\left(\frac{\partial V}{\partial W}\right)^2 \sigma^2}$$

$$\sigma_v = \sqrt{2(.62892 \text{ in})^2 (.0001 \text{ in})^2 + 2(4.99 \cdot 10^{-6} \text{ in})^2 (.0001 \text{ in.})^2 + 2(0.00175 \text{ in})^2 (.005 \text{ in})^2} = .000889 \text{ in}^3$$

Does this satisfy the criterion that the derived measurement uncertainty be 5% or less of the derived measurement?

$$\frac{.000889}{.0315} = .02823(100\%) = 2.823\%$$

Ans: _____yes_____

(L,T: caliper), (W:micrometer) Perform the same calculation using the caliper measurements for L, micrometer measurements for W, and the caliper measurements for T. Again, recalculate the numerical values for the partial derivatives – show these. Show the formula for σ_v and then its numerical evaluation.

$$\sqrt{2(.62892in)^2(.0001in)^2 + 2(.63in)^2(.005in)^2 + 2(.00175in)^2(.005in)^2} = \sigma_v$$

$$= 4.454 \cdot 10^{-3} in^3$$

$$\frac{4.454 \cdot 10^{-3}}{.0315} = .1414219\%$$

$$= 14.142\%$$

Does this satisfy the criterion that the derived measurement uncertainty be 5% or less of the derived measurement?

Ans: _____No_____

(L,W: caliper), (T:micrometer) Finally, find σ_v with the caliper measurements for L, caliper measurements for W, and micrometer measurements for T. Again, recalculate the numerical values for the partial derivatives – show these. Show the formula for σ_v and then its numerical evaluation.

$$\sigma_v = \sqrt{2(.00175)^2(.005)^2 + 2(0.009)^2(.005)^2 + 2(4.99 \cdot 10^{-6})^2(.0001)^2}$$

$$= 6.483 \cdot 10^{-5} in^3$$

$$6.483 \cdot 10^{-5} \frac{1}{.0315}$$

$$\sigma = 0.2058\%$$

Does this satisfy the criterion that the derived measurement uncertainty be 5% or less of the derived measurement?

Ans: __yes__

In conclusion, restate the volume and uncertainty in volume you found for each combination as $V \pm \sigma_V$, where σ_V is shown as a percentage of V . What combination of measurements should a worker use to minimize workload while still satisfying the requirement that the uncertainty in the measured volume must be no greater than 5%?

Uncertainties for Cases 1-4 Results:

1	14.12%
2	2.823%
3	14.142%
4	0.2058%

For our group, the cases that had low uncertainties and that were less than 5%, were cases 2 and 4, therefore, the combination of measurements that should therefore be used to minimize the workload and still having a low uncertainty(<5%) would be using the micrometer to measure the width and thickness, while using the calipers to measure the length. The combinations 2 and 4, yielding the best results were due to using the micrometer, which is much more precise.